

Student Lifestyle, Mental Health and Burnout Dataset Analysis

Multivariate Analysis – Project

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1. Introduction

Student mental health has become an increasingly important issue in modern educational environments due to growing academic demands, digital exposure, financial pressure, and social expectations. Variables such as stress, anxiety, depression, burnout, sleep patterns, physical activity, and social support all contribute to students' overall well-being and academic performance. Understanding the relationships among these variables is useful for identifying dominant patterns, latent structures, and risk categories.

This study applies multivariate statistical techniques to a large student mental health and burnout dataset in order to extract meaningful patterns and provide interpretable insights. In particular, Principal Component Analysis (PCA), Factor Analysis (FA), and Quadratic Discriminant Analysis (QDA) are used to examine dimensionality reduction, latent factor structure, and classification of student risk levels, respectively.

1.1. Objectives

- To analyze the student lifestyle, mental health and burnout dataset using multivariate statistical techniques.
- To reduce dimensionality using Principal Component Analysis (PCA).
- To identify latent structures using Factor Analysis (FA).
- To classify students into risk groups using Quadratic Discriminant Analysis (QDA).

1.2. Main Question

Can the dimensionality of the dataset be reduced while preserving important variability, and can meaningful latent structures and risk classifications be identified from the academic, psychological, and lifestyle-related variables?

1.3. Purpose

To perform the following multivariate techniques:

- Principal Component Analysis (PCA)
- Factor Analysis (FA)
- Quadratic Discriminant Analysis (QDA)

2. Methodology

2.1. Dataset Description

The dataset used in this study is obtained from Kaggle : <https://www.kaggle.com/datasets/ayeshasiddiqua123/student-health>.

This contains **1,000,000 observations** and **20 features**, making it suitable for large-scale multivariate analysis. Each observation represents an individual student, and the variables describe demographic, academic, psychological, behavioural, and lifestyle-related characteristics.

The features include demographic variables such as age, gender, and academic year; academic variables such as study hours per day, exam pressure, and academic performance; psychological indicators such as stress level, anxiety score, depression score, burnout score, and mental health index; lifestyle-related variables such as sleep hours, physical activity, screen time, and internet usage; and external or contextual factors such as social support, financial stress, family expectation, dropout risk, and risk level.

The target variable used in classification was **risk_level**, with categories **High**, **Medium**, and **Low**.

2.2. Data Preprocessing

Before applying the multivariate methods, the dataset was preprocessed to ensure quality and suitability for analysis.

2.2.1. Handling Missing Values

All rows containing missing values were removed using listwise deletion so that only complete observations remained.

2.2.2. Removing Outcome-Type Variables

The variables `mental_health_index` and `dropout_risk` were removed at the beginning of the analysis because they behave as downstream or outcome-related variables and could distort interpretation in the unsupervised and supervised analyses.

2.2.3. Removal of Categorical Variables

The variables `age` and `academic_year` were removed from the dataset as they are discrete/categorical in nature. Techniques such as PCA and Factor Analysis assume contin-

uous variables and rely on correlation structures. Including such variables could distort the covariance matrix and lead to misleading components and factors.

2.2.4. Outlier Detection and Removal

Outliers were detected using Z-scores computed from the numerical variables. Observations with any standardized value exceeding ± 3 were removed from the analysis.

2.2.5. Feature Scaling and Target Separation

All numerical variables were standardized before PCA, FA and QDA to ensure comparability across different measurement scales. Without standardization, variables with larger magnitudes could dominate the analysis.

For PCA and FA, only numerical predictors were used. After preprocessing, a total of **14 predictor variables** were retained for these methods. For QDA, the predictors were the numerical variables and the target variable was `risk_level`, resulting in a total of **15 variables**.

2.3. Methods Employed

2.3.1. Principal Component Analysis (PCA)

PCA was used to reduce dimensionality and to identify the principal directions of variability (Jolliffe, 2025) in the dataset. The number of principal components was evaluated using eigenvalues, cumulative variance explained, and the scree plot.

2.3.2. Factor Analysis (FA)

Factor Analysis was used to identify latent factors that explain the observed correlation structure (Yong et al., 2013) among the variables. Sampling adequacy was checked using the KMO measure and Bartlett's test of sphericity. Parallel analysis was used to guide the number of retained factors.

2.3.3. Quadratic Discriminant Analysis (QDA)

QDA was applied as a supervised classification technique (Tharwat, 2016) to classify students into **High**, **Medium**, and **Low** risk groups. Unlike LDA, QDA allows each class to have its own covariance matrix, making it more suitable when the assumption of equal covariance matrices is violated.

3. Results and Discussion

3.1. Principal Component Analysis

Principal Component Analysis was performed on the standardized numerical variables to reduce dimensionality and identify the dominant patterns in the data. The correlation matrix showed several meaningful relationships among the variables. In particular, stress level, anxiety score, depression score, and burnout score were positively correlated, indicating a common psychological dimension. Study hours and exam pressure were also positively related, reflecting academic workload, while screen time and internet usage were very strongly correlated, suggesting a shared behavioral pattern.

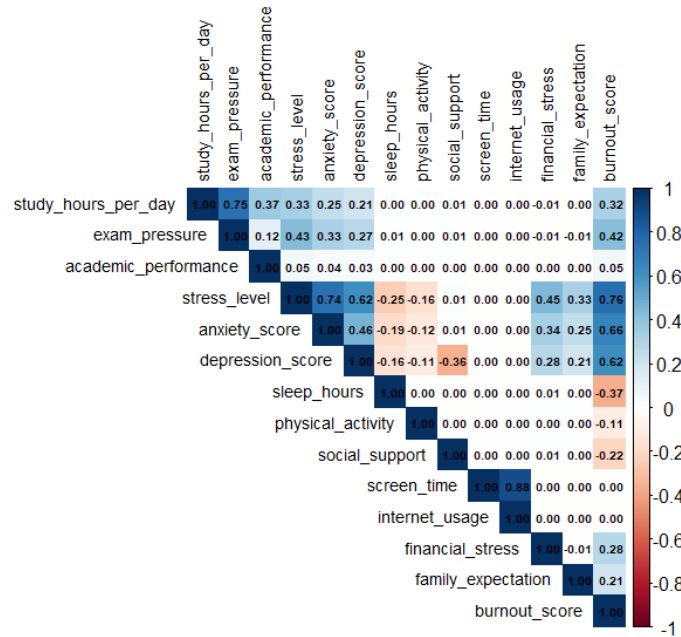


Figure 1: Correlation matrix of variables used in PCA

The PCA results showed that the first principal component explained **26.9%** of the total variance. The first two components explained **40.4%**, while the first **7** components explained approximately **82.6%** of the total variance. Around **89.1%** of the variance was explained by the first 8 components. Based on the Kaiser criterion, **7 principal components** had eigenvalues greater than one. The scree plot showed a gradual decline with a mild elbow around the third to fourth component, indicating that the dataset is influenced by multiple dimensions rather than a very small number of dominant components.

Table 1: Eigenvalues and explained variance from PCA

Component	Eigenvalue	Variance (%)	Cumulative Variance (%)
PC1	3.7714	26.9	26.9
PC2	1.8889	13.5	40.4
PC3	1.6653	11.9	52.3
PC4	1.1841	8.5	60.8
PC5	1.0391	7.4	68.2
PC6	1.0073	7.2	75.4
PC7	1.0016	7.2	82.6

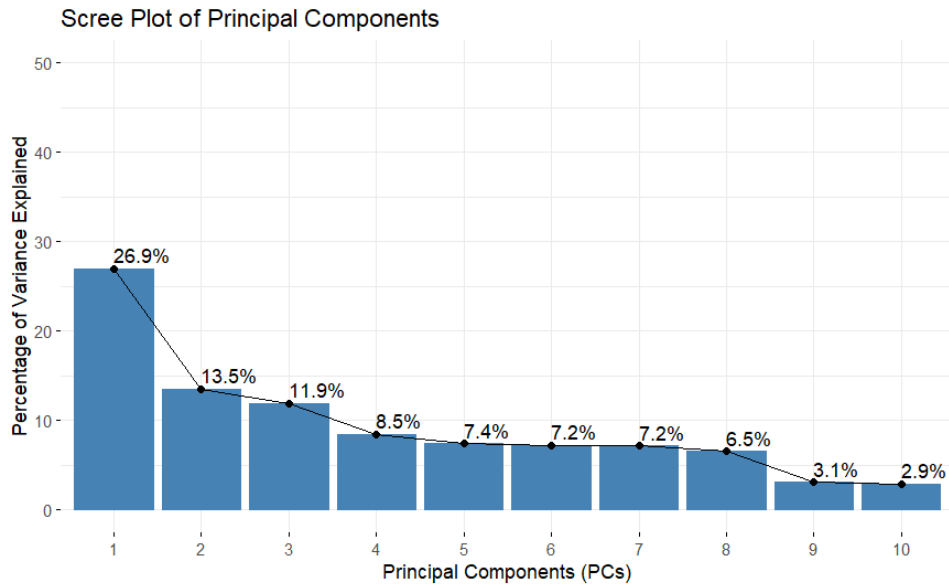


Figure 2: Scree plot of principal components

The first principal component (PC1) was mainly driven by stress level, burnout score, anxiety score, and depression score, and can therefore be interpreted as a **psychological distress component**. The second principal component (PC2) was dominated by screen time and internet usage, and represents a **digital behavior component**. The third principal component (PC3) was associated with study hours per day, exam pressure, and academic performance, reflecting an **academic pressure component**. The fourth component was strongly associated with social support, while the fifth component was primarily driven by sleep hours, capturing **social support** and **sleep/well-being** dimensions.

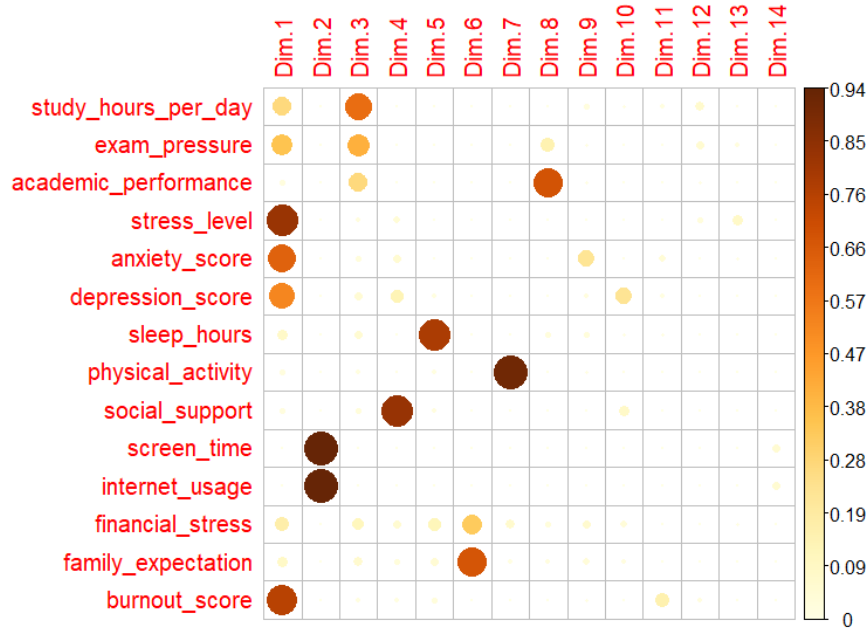


Figure 3: Quality of representation ($\cos^2$) of variables across principal components

3.2. Factor Analysis

Factor Analysis was carried out to uncover latent factors underlying the student lifestyle, mental health and burnout variables. The KMO measure of sampling adequacy was 0.65, indicating acceptable adequacy for factor analysis. Bartlett's test of sphericity was highly significant ($p < 0.001$), showing that the variables were sufficiently correlated to justify factor extraction.

Parallel analysis suggested that **7** factors should be retained, and therefore a **7 factor model** was fitted using maximum likelihood extraction with varimax rotation. The model explained approximately **61.09%** of the total variance. The first six factors contributed the majority of the meaningful structure, while the seventh factor had very low explanatory power.

Table 2: Variance accounted for by the 7-factor model

	ML1	ML2	ML3	ML4	ML7	ML6	ML5
SS loadings	2.0794	1.8839	1.8283	0.9977	0.8916	0.7664	0.1061
Proportion Var	0.1485	0.1346	0.1306	0.0713	0.0637	0.0547	0.0076
Cumulative Var	0.1485	0.2831	0.4137	0.4849	0.5486	0.6034	0.6109

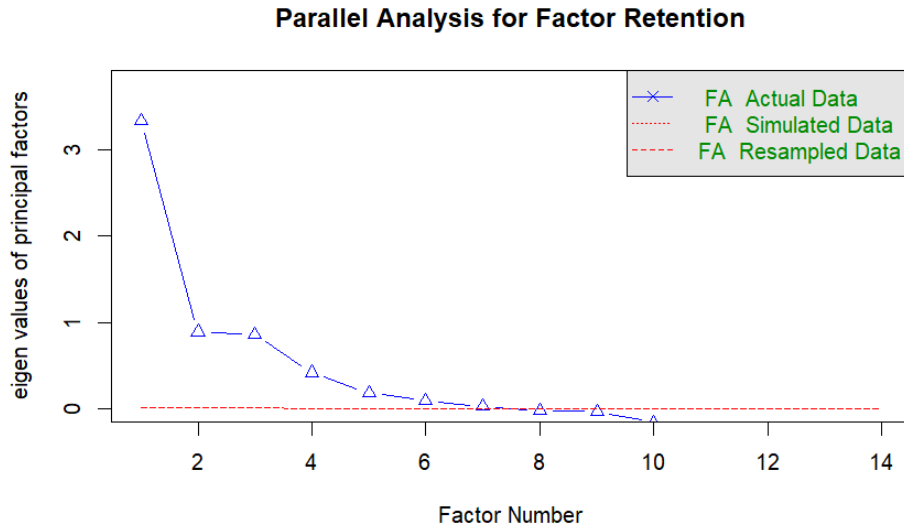


Figure 4: Parallel analysis for factor retention

The rotated factor loadings revealed several interpretable dimensions. **Factor ML1** was characterized by stress level, anxiety score, financial stress, depression score, and burnout score, representing a **psychological and financial distress factor**. **Factor ML2** was dominated by screen time and internet usage, representing a **digital behavior factor**. **Factor ML3** was defined mainly by exam pressure and study hours per day, representing an **academic pressure factor**. **Factor ML4** was characterized by burnout score and sleep hours, representing a **burnout and recovery factor**. **Factor ML7** was associated with social support and depression score, suggesting a **social-emotional adjustment factor**. **Factor ML6** loaded on study hours and academic performance, indicating an **academic performance and effort factor**.

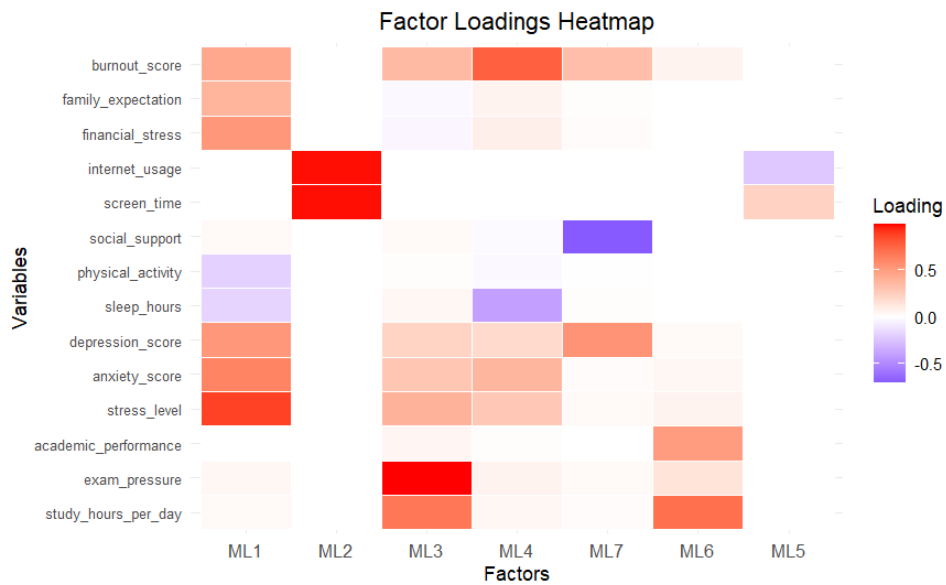


Figure 5: Factor loadings heatmap

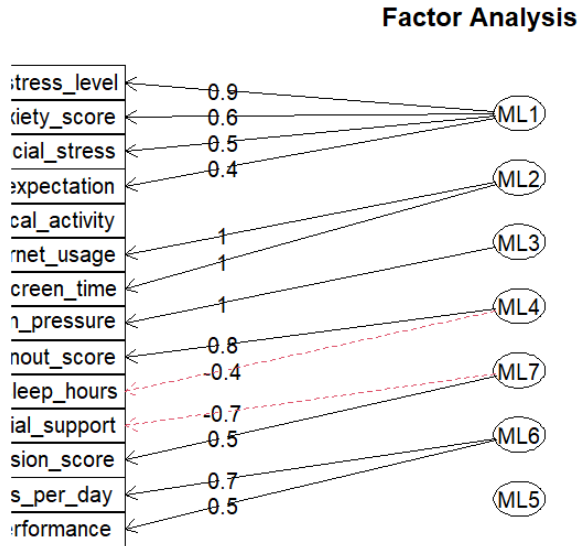


Figure 6: Rotated factor structure diagram

Table 3: Strong factor loadings for the rotated 7-factor solution

Factor	Variable	Loading
ML1	stress_level	0.8720
ML1	anxiety_score	0.6123
ML1	financial_stress	0.5177
ML1	depression_score	0.5170
ML1	burnout_score	0.4373
ML2	screen_time	0.9705
ML2	internet_usage	0.9705
ML3	exam_pressure	0.9843
ML3	study_hours_per_day	0.6687
ML4	burnout_score	0.7569
ML4	sleep_hours	-0.4021
ML7	social_support	-0.7040
ML7	depression_score	0.5354
ML6	study_hours_per_day	0.6976
ML6	academic_performance	0.4996

3.3. Quadratic Discriminant Analysis Results

Quadratic Discriminant Analysis (QDA) was applied as an alternative to LDA in order to account for unequal covariance structures across risk groups. After preprocessing, the dataset consisted of 979,254 observations and **15 variables** (including the response

variable), with **14 standardized predictor variables**. The class distribution was highly imbalanced, with 77.51% of observations in the low-risk group, 21.66% in the medium-risk group, and only 0.82% in the high-risk group.

Table 4: Prior probabilities of risk groups (QDA)

Risk Level	Prior Probability
High	0.0082
Low	0.7751
Medium	0.2166

The assumption of equal covariance matrices was formally tested using Box’s M test. The test was highly significant ($\chi^2 = 73714$, $df = 210$, $p < 0.001$), indicating that covariance matrices differ significantly across groups. This confirms that QDA is more appropriate than LDA for this dataset.

To evaluate predictive performance, cross-validation was applied. The QDA model achieved a high overall classification accuracy of **98.53%**. However, due to the strong class imbalance, class-wise performance metrics were also examined.

Table 5: Confusion matrix for QDA classification

Predicted / Actual	High	Low	Medium
High	8014	0	466
Low	0	749066	3917
Medium	39	10002	207750

Table 6: Class-wise performance metrics (QDA)

Class	Precision	Recall	F1-score
High	0.9450	0.9952	0.9695
Low	0.9948	0.9868	0.9908
Medium	0.9539	0.9793	0.9664

The balanced accuracy of the model was **0.9871**, indicating consistently strong performance across all classes, including the minority high-risk group. This shows that the model performs well not only for the majority class but also for the smaller and more critical minority class.

Overall, the QDA results demonstrate that student risk level can be classified very effectively using psychological, academic, and lifestyle variables. The strong class-wise

performance confirms that QDA captures important nonlinear relationships among the predictors and provides a flexible classification framework for the dataset.

4. Conclusion and Recommendations

This study applied Principal Component Analysis, Factor Analysis, and Quadratic Discriminant Analysis to examine student mental health and burnout. The results consistently indicate that psychological distress is the most influential factor shaping student outcomes.

PCA showed that variability in the data is explained by multiple dimensions, including psychological distress, academic pressure, digital behavior, and lifestyle factors. Factor Analysis supported these findings by identifying similar latent structures such as psychological distress, academic pressure, burnout and recovery, and social-emotional adjustment.

QDA achieved high classification performance, demonstrating that student risk levels can be effectively predicted using psychological and lifestyle variables. Key factors influencing risk included burnout, stress, anxiety, sleep, and social support.

PCA and Factor Analysis provided consistent insights into the data structure, while QDA complemented them by offering strong predictive capability. Together, these methods provide a comprehensive understanding of the dataset by combining dimensionality reduction, latent structure identification, and classification.

Overall, the results suggest that student burnout and mental health are multidimensional and are influenced by emotional, academic, behavioral, and social factors.

4.1. Limitations

- The dataset is highly imbalanced, with a very small proportion of high-risk students.
- Multicollinearity exists among several predictors, particularly among psychological variables such as stress, anxiety, depression, and burnout.
- Some variables had low communalities in the factor analysis, indicating weaker representation by the extracted factors.
- The interpretation of principal components and factors is partly subjective and depends on loading patterns and rotation methods.
- The factor model showed moderate fit (e.g., RMSEA), suggesting that not all variability is fully captured.

4.2. Recommendations

- Techniques to address class imbalance, such as resampling or weighting methods, can be explored to further improve classification performance.
- Additional analyses may investigate longitudinal or time-dependent aspects of student mental health.
- Variables related to stress, burnout, sleep, and social support should be prioritized in predictive modeling.
- Future studies may consider incorporating additional behavioral or environmental variables to improve model interpretability and predictive power.

References

- Jolliffe, I. (2025). Principal component analysis. In *International Encyclopedia of Statistical Science*, pages 1945–1948. Springer.
- Tharwat, A. (2016). Linear vs. quadratic discriminant analysis classifier: a tutorial. *International Journal of Applied Pattern Recognition*, 3(2):145–180.
- Yong, A. G., Pearce, S., et al. (2013). A beginner’s guide to factor analysis: Focusing on exploratory factor analysis. *Tutorials in quantitative methods for psychology*, 9(2):79–94.

5. Appendices

```
1 # DATA LOADING & PREPROCESSING
2
3 data <- read.csv("student_mental_health_burnout_1M.csv")
4
5 set.seed(123)
6
7 # Convert categorical variables
8 data$gender <- as.factor(data$gender)
9 data$academic_year <- as.factor(data$academic_year)
10 data$risk_level <- as.factor(data$risk_level)
11
12 # Remove missing values
13 data <- na.omit(data)
14
15 # Remove outcome / derived and Discrete variables
16 remove_vars <- c("mental_health_index", "dropout_risk", "age")
17 remove_vars <- remove_vars[remove_vars %in% colnames(data)]
18 data <- data[, !(colnames(data) %in% remove_vars)]
19
20 # Numeric subset
21 student_num <- data[, sapply(data, is.numeric), drop = FALSE]
22
23 # Outlier detection using numeric variables
24 z_scores <- scale(student_num)
25 outlier_rows <- apply(abs(z_scores) > 3, 1, any)
26 outlier_count <- sum(outlier_rows)
27 cat("Number of outlier rows:", outlier_count, "\n")
28
29 # Remove outliers from BOTH full data and numeric data
30 data <- data[!outlier_rows, ]
31 student_num <- data[, sapply(data, is.numeric), drop = FALSE]
```

Listing 1: R Code for Data Loading and Preprocessing

```
1 # PRINCIPAL COMPONENT ANALYSIS (PCA)
2
3 # Correlation matrix
4 cor_matrix <- cor(student_num, method = "spearman")
5
6 cat("\nCorrelation Matrix:\n")
7 print(round(cor_matrix, 2))
```

```

8
9 corrplot(cor_matrix, method = "color", type = "upper", tl.col = "
  black", tl.cex = 0.7, addCoef.col = "black", number.cex = 0.5)
10
11 # Apply PCA
12 pca_result <- prcomp(student_num, scale. = TRUE)
13
14 cat("\nObjects in PCA result:\n")
15 print(names(pca_result))
16
17 cat("\nVariable means used for centering:\n")
18 print(pca_result$center)
19
20 cat("\nVariable scales used for standardization:\n")
21 print(pca_result$scale)
22
23 # PCA loadings
24
25 cat("\nPCA Loadings:\n")
26 print(round(pca_result$rotation, 4))
27
28 # PCA scores
29
30 cat("\nFirst 6 PCA scores:\n")
31 print(head(pca_result$x))
32
33 # Standard deviations, eigenvalues
34
35 cat("\nStandard deviations of PCs:\n")
36 print(pca_result$sdev)
37
38 VE <- pca_result$sdev^2
39 cat("\nEigenvalues:\n")
40 print(VE)
41
42 PVE <- VE / sum(VE)
43 cat("\nProportion of variance explained:\n")
44 print(round(PVE, 3))
45
46 cat("\nCumulative variance explained:\n")
47 print(round(cumsum(PVE), 3))
48
49 cat("\nPCs with eigenvalue > 1 (Kaiser criterion):\n")

```

```

50 print(which(VE > 1))
51
52 # Eigenvalue table
53
54 eig.val <- get_eigenvalue(pca_result)
55 cat("\nEigenvalue Table:\n")
56 print(eig.val)
57
58 # Scree plot
59
60 fviz_eig(pca_result,addlabels = TRUE,ylim = c(0, 50),xlab = "
    Principal Components (PCs)",ylab = "Percentage of Variance
    Explained",main = "Scree Plot of Principal Components")
61
62 # Quality of representation
63
64 var <- get_pca_var(pca_result)
65
66 cat("\nQuality of representation:\n")
67 print(head(var$cos2, ncol(student_num)))
68
69 corrplot(var$cos2,is.corr = FALSE)
70
71 # Top contributing variables to first 7 PCs
72
73 loadings_df <- as.data.frame(pca_result$rotation)
74
75 for (pc in colnames(loadings_df)[1:min(7, ncol(loadings_df))]) {
76   cat("\nTop variables for", pc, ":\n")
77   temp <- data.frame(
78     Variable = rownames(loadings_df),
79     Loading = loadings_df[[pc]]
80   )
81   temp <- temp[order(abs(temp$Loading), decreasing = TRUE), ]
82   temp$Loading <- round(temp$Loading, 4)
83   print(head(temp, 8))
84 }

```

Listing 2: R code for Principal Component Analysis

```

1
2 # FACTOR ANALYSIS (FA)
3
4 # FA uses the same student_num

```



```

5 fa_data <- student_num
6
7 # KMO test
8
9 cat("\nKMO Test:\n")
10 kmo_result <- KMO(fa_data)
11 print(kmo_result)
12
13 # Bartlett's test
14
15 cat("\nBartlett Test:\n")
16 bartlett_result <- cortest.bartlett(cor(fa_data), n = nrow(fa_
    data))
17 print(bartlett_result)
18
19 # Parallel analysis
20
21 fa.parallel(fa_data, fa = "fa", main = "Parallel Analysis for
    Factor Retention")
22
23 # Fit factor model
24
25 fa_model <- fa(fa_data, nfactors = 7, rotate = "varimax", fm = "ml")
26
27 cat("\nFactor Analysis Summary:\n")
28 print(fa_model)
29
30 # Factor loadings
31
32 loadings_df <- as.data.frame(unclass(fa_model$loadings))
33 cat("\nFactor Loadings:\n")
34 print(round(loadings_df, 4))
35
36 # Top variables per factor
37
38 result <- data.frame()
39
40 for (f in colnames(loadings_df)) {
41   temp <- data.frame(
42     Factor = f,
43     Variable = rownames(loadings_df),
44     Loading = loadings_df[[f]]
45   )

```

```

46   temp <- temp[abs(temp$Loading) > 0.4, ]
47   temp <- temp[order(abs(temp$Loading), decreasing = TRUE), ]
48   result <- rbind(result, temp)
49 }
50
51 result$Loading <- round(result$Loading, 4)
52
53 cat("\nTop Variables per Factor (|loading| > 0.4):\n")
54 print(result)
55
56 # Communalities
57
58 cat("\nCommunalities:\n")
59 print(round(fa_model$communality, 4))
60
61 # Variance accounted
62
63 cat("\nVariance Accounted For:\n")
64 print(round(fa_model$Vaccounted, 4))
65
66 # Factor diagram
67
68 fa.diagram(fa_model)
69
70 # Factor loadings heatmap
71
72 load_matrix <- round(as.matrix(loadings_df), 2)
73
74 load_df <- melt(load_matrix)
75 colnames(load_df) <- c("Variable", "Factor", "Loading")
76
77 ggplot(load_df, aes(x = Factor, y = Variable, fill = Loading)) +
78   geom_tile(color = "white") +
79   scale_fill_gradient2(low = "blue", mid = "white", high = "red",
80     midpoint = 0) +
81   theme_minimal() +
82   labs(title = "Factor Loadings Heatmap",
83     x = "Factors",
84     y = "Variables") +
85   theme(
86     axis.text.y = element_text(size = 8),
87     axis.text.x = element_text(angle = 0, size = 10),
88     plot.title = element_text(hjust = 0.5)

```

```

88   )
89
90   library(dplyr)
91   library(knitr)
92   library(kableExtra)
93
94   result_clean <- result %>%
95     arrange(Factor, desc(abs>Loading)))
96
97   kable(result_clean,
98     caption = "Strong factor loadings for the rotated 7-factor
99     solution",
100     digits = 4) %>%
101     kable_styling(full_width = FALSE)

```

Listing 3: R code for Factor Analysis

```

1  # QUADRATIC DISCRIMINANT ANALYSIS (QDA)
2
3  qda_data <- data
4  qda_data$risk_level <- as.factor(qda_data$risk_level)
5
6  # Numeric predictors
7  qda_num <- qda_data[, sapply(qda_data, is.numeric), drop = FALSE]
8
9  # Standardize predictors
10 qda_num <- scale(qda_num)
11
12 qda_df <- data.frame(qda_num, risk_level = qda_data$risk_level)
13
14 cat("\nDimensions of QDA dataset:\n")
15 print(dim(qda_df))
16
17 cat("\nClass distribution:\n")
18 print(table(qda_df$risk_level))
19
20
21 # Assumption check : Equality of covariance matrices
22
23 library(biotools)
24
25 boxm_result <- boxM(qda_num, qda_df$risk_level)
26
27 cat("\nBox's M Test for Homogeneity of Covariance Matrices:\n")

```

```

28 print(boxm_result)
29
30
31 # Fit QDA model
32
33 library(MASS)
34
35 qda_model <- qda(risk_level ~ ., data = qda_df)
36
37 cat("\nQDA Model:\n")
38 print(qda_model)
39
40 # Prior probabilities
41
42 cat("\nPrior probabilities of groups:\n")
43 print(round(qda_model$prior, 4))
44
45 # Cross validation
46
47 qda_cv <- qda(risk_level ~ ., data = qda_df, CV = TRUE)
48
49 # Confusion matrix
50
51 conf_mat_qda <- table(Predicted = qda_cv$class, Actual = qda_df$
    risk_level)
52
53 cat("\nConfusion Matrix:\n")
54 print(conf_mat_qda)
55
56 # Overall accuracy
57
58 qda_accuracy <- mean(qda_cv$class == qda_df$risk_level)
59 cat("\nQDA Cross-validated Accuracy:", round(qda_accuracy, 4), "\
    n")
60
61 # Class-wise metrics
62
63 actual <- qda_df$risk_level
64 pred <- qda_cv$class
65 cm <- conf_mat_qda
66
67 classes <- levels(actual)
68

```

```

69 metrics <- data.frame(Class = classes, Precision = NA, Recall = NA,
70   F1 = NA)
71
72 for (i in seq_along(classes)) {
73   cls <- classes[i]
74
75   tp <- cm[cls, cls]
76   fp <- sum(cm[cls, ]) - tp
77   fn <- sum(cm[, cls]) - tp
78
79   precision <- ifelse((tp + fp) == 0, NA, tp / (tp + fp))
80   recall <- ifelse((tp + fn) == 0, NA, tp / (tp + fn))
81   f1 <- ifelse(is.na(precision) | is.na(recall) | (
82     precision + recall) == 0,
83     NA,
84     2 * precision * recall / (precision +
85       recall))
86
87   metrics$Precision[i] <- precision
88   metrics$Recall[i] <- recall
89   metrics$F1[i] <- f1
90 }
91
92 metrics[, 2:4] <- round(metrics[, 2:4], 4)
93
94 cat("\nClass-wise Precision, Recall, and F1:\n")
95 print(metrics)
96
97 # Balanced accuracy
98
99 balanced_accuracy <- mean(metrics$Recall, na.rm = TRUE)
100 cat("\nBalanced Accuracy:", round(balanced_accuracy, 4), "\n")

```

Listing 4: R code for Quadratic Discriminant Analysis